Design Report - Single Stage CMOS Differential Amplifier

Abstract

This project presents the design and verification of a single-stage CMOS differential amplifier with a PMOS current mirror active load using the GPDK180 (180 nm) CMOS process. The amplifier was designed through analytical transistor sizing and verified using Cadence Virtuoso. This report summarizes the design methodology, key engineering decisions, and verification results. Complete design calculations are available in the accompanying hand calculations document.

hand_calculations.pdf

1. Design Methodology

The amplifier was designed using an analytical approach before circuit implementation. The design process consisted of:

- Determining the bias current from the slew-rate specification.
- Selecting appropriate transistor overdrive voltages.
- Calculating device dimensions using long-channel MOSFET equations.
- Verifying transistor saturation conditions.
- Estimating the small-signal voltage gain.
- Implementing and validating the design in Cadence Virtuoso through DC operating point and AC analyses.

Detailed derivations and transistor sizing calculations are provided in

hand_calculations.pdf

2. Design Decisions

Channel Length Selection

Although the minimum channel length supported by GPDK180 is **180 nm**, a channel length of **800 nm** was selected for all transistors. This choice reduces channel-length modulation and increases the intrinsic output resistance of the devices, resulting in higher voltage gain. At the same time, it maintains acceptable parasitic capacitances, providing a practical balance between gain and bandwidth.

PMOS Bias Selection

The PMOS current mirror was designed with **VSG = 0.60 V** instead of the calculated value of approximately **0.68 V**. This provides additional saturation margin and prevents operation near the edge of saturation, improving robustness against output voltage variations while maintaining the desired bias current.

3. Design Verification

The completed circuit was verified using **Cadence Virtuoso** with the **Spectre** simulator.

The following analyses were performed:

- DC Operating Point Analysis
- AC Frequency Response Analysis

The simulation results confirmed proper device biasing and validated the analytical design procedure. Circuit schematics and simulation plots are available in the repository's **Images/** directory.

4. Results

The completed design was verified using **Cadence Virtuoso** with the **Spectre** simulator. DC operating point analysis confirmed that all transistors operated in saturation, while AC analysis validated the small-signal performance of the amplifier. The circuit schematics and simulation plots are provided in the repository's **Images/** directory.

Parameter	**Specification**	**Simulated**
Voltage Gain	≥ 40 dB	**38.963 dB**
Unity Gain Bandwidth	—	**338.844 MHz**
−3 dB Bandwidth	≥ 5 MHz	**5.472 MHz**

The simulated voltage gain is slightly below the target specification of **40 dB**, with a deviation of approximately **1.04 dB**. However, the amplifier satisfies the bandwidth requirement, achieving a **−3 dB bandwidth of 5.472 MHz** and a **unity gain bandwidth of 338.844 MHz**. The slight reduction in gain can be attributed to second-order effects, including channel-length modulation, finite output resistance, and parasitic capacitances accounted for in the BSIM device models.

5. Conclusion

A single-stage CMOS differential amplifier with a PMOS current mirror active load was successfully designed and verified using Cadence Virtuoso. The design achieved a simulated voltage gain of **38.963 dB**, a **−3 dB bandwidth of 5.472 MHz**, and a **unity gain bandwidth of 338.844 MHz**. Although the gain is slightly below the target specification, the amplifier satisfies the required bandwidth and demonstrates a systematic analog IC design methodology from analytical sizing to simulation-based verification. Further optimization of the transistor dimensions, particularly the **channel length**, could improve the voltage gain while maintaining the desired bandwidth.